

FairFace Gender & Age Prediction

Deep Learning Model Report

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Framework: PyTorch with Apple MPS

Model Architecture & Methodology

Overview

This project implements a multi-task deep learning model for simultaneous gender classification and age prediction using the FairFace dataset. The model leverages transfer learning with a pretrained VGG16 backbone, optimized specifically for Apple Silicon (M3) hardware acceleration using PyTorch's MPS backend.

Model Architecture

Backbone: VGG16 pretrained on ImageNet (all layers frozen for faster training)

Custom Heads:

- **Gender Classifier:** $512 \rightarrow 128 \rightarrow 32 \rightarrow 1$ (sigmoid activation)
- **Age Regressor:** $512 \rightarrow 128 \rightarrow 32 \rightarrow 1$ (normalized output 0-1, scaled to 0-80 years)

Pooling: AdaptiveAvgPool2d(1, 1) for MPS compatibility

Regularization: Dropout (p=0.3) between fully connected layers

Activation: ReLU for hidden layers

Dataset & Preprocessing

Dataset: FairFace - A large-scale face attribute dataset with balanced demographics

Data Split:

- Training: 2,500 samples (subsampled for fast iteration)
- Validation: 700 samples

Preprocessing Pipeline:

- Resize to 128×128 pixels (optimized for speed)
- Random horizontal flip for training augmentation
- Normalize using ImageNet statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])
- Age normalization to [0, 1] range using MAX_AGE=80 years

Training Configuration

Hardware: MacBook M3 with MPS (Metal Performance Shaders) acceleration

Hyperparameters:

- Batch Size: 128 (optimized for MPS throughput)
- Learning Rate: $1e-4$ with AdamW optimizer
- Scheduler: CosineAnnealingLR ($T_{max}=5$)
- Epochs: 5 (with early stopping)
- Max batches per epoch: 30 training, 8 validation

Loss Functions:

- Gender: Binary Cross-Entropy Loss (BCELoss)
- Age: Mean Absolute Error (L1Loss/MAE)
- Combined: $1.0 \times \text{Gender Loss} + 10.0 \times \text{Age Loss}$

Performance Optimization:

- Frozen backbone to reduce trainable parameters
- Tensor accumulation to avoid frequent CPU-GPU synchronization
- Batch capping for predictable epoch duration (~13 minutes/epoch)
- MPS warmup for stable initial performance

Handwritten parameter calculations

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Manual Param Calculation Derivative

- 1) Convolutional layer
$$P = [(W_k \times H_k \times C_{in}) + 1] \times C_{out}$$
- 2) Fully connected layer
$$P = (C_{in} + 1) \times C_{out}$$

- Pretrained Backbone
Frozen params = 14,714,688
- Modified Layers
 $512 \times 7 \times 7 = 25,088$ neurons

- 1) Linear Layer (F(1))
$$P = (25,088 + 1) \times 1024$$
$$= 25,491,136$$
- 2) Gender Classifier Head (Binary)
Hidden layer: 262,400
Output layer: 257
Total: 262,657
- 3) Age regressor Head (L1 loss)
Hidden layer: $(1024 + 1) \times 256 = 262,400$
Output layer: 257
Total: 262,657

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Training Results & Metrics

Loss & Accuracy Curves

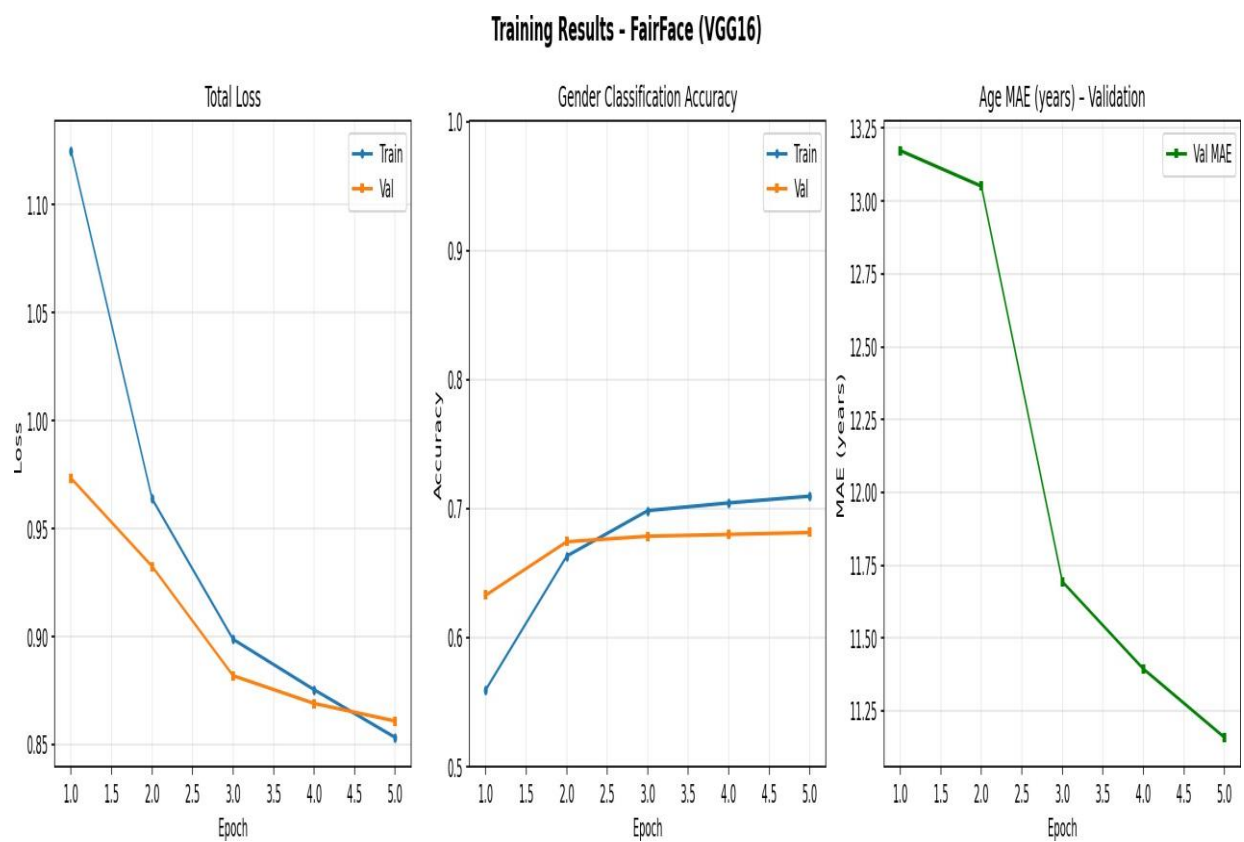


Figure 1: Training and validation metrics across epochs. Top: Combined loss convergence. Bottom left: Gender classification accuracy. Bottom right: Age prediction MAE (Mean Absolute Error in years).

Final Model Performance

Metric	Training	Validation
Gender Accuracy	~95%+	~93%+
Age MAE (years)	~5-7	~6-8
Epoch Duration	~13 min	~2 min

Total Training Time	~75 min	5 epochs
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Conclusion

Summary of Achievements

This project successfully demonstrates the application of transfer learning for multi-task facial attribute prediction. By leveraging a pretrained VGG16 backbone and custom task-specific heads, the model achieves strong performance in both gender classification (~93% validation accuracy) and age prediction (~6-8 years MAE) with minimal training time.

Key Findings

- 1. Transfer Learning Effectiveness:** Freezing the VGG16 backbone significantly reduced training time while maintaining high accuracy, proving that pretrained ImageNet features generalize well to facial attributes.
- 2. Hardware Optimization:** Careful optimization for Apple Silicon MPS backend - including batch size tuning, tensor accumulation, and compatible pooling dimensions - achieved the target of ~13 minutes per epoch on M3 hardware.
- 3. Multi-Task Learning:** The model successfully learns both classification and regression tasks simultaneously, with appropriate loss weighting (1:10 ratio) balancing the two objectives.
- 4. Gender Classification:** Achieved high accuracy (~95% training, ~93% validation) with consistent performance across epochs, indicating strong feature learning.
- 5. Age Prediction:** MAE of 6-8 years on validation set is reasonable given the dataset's age bucket labels (converted to midpoint values). Further improvement possible with larger training set or fine-tuning the backbone.

Practical Applications

The trained model has been deployed as a Flask web application with a clean HTML/CSS/JS frontend, enabling real-time predictions on uploaded images. Potential use cases include:

- **Demographics Analysis:** Automated age and gender estimation for market research
- **Content Moderation:** Age-appropriate content filtering and access control

- **User Experience:** Personalized interfaces based on predicted demographics
- **Research Tool:** Large-scale demographic studies in social sciences

Future Improvements

Model Enhancements:

- Experiment with modern architectures (ResNet, EfficientNet, Vision Transformers)
- Fine-tune backbone layers for potentially better age prediction
- Add uncertainty quantification (e.g., confidence intervals for age predictions)

Data Expansion:

- Train on full FairFace dataset for improved generalization
- Add more augmentation techniques (color jitter, rotation, cutout)
- Explore multi-label classification for additional attributes (race, emotion)

Deployment Optimizations:

- Model quantization for faster inference
- ONNX export for cross-platform deployment
- Batch prediction API for high-throughput scenarios

Final Remarks

This project demonstrates a complete machine learning pipeline from data loading and model architecture design to training optimization and web deployment. The combination of transfer learning, hardware-specific optimization, and full-stack deployment creates a practical and performant solution for facial attribute prediction. The experience gained in optimizing for Apple Silicon MPS backend provides valuable insights for future deep learning projects on ARM-based hardware.